

Counting Divisors

Time limit: 1.00 s

Memory limit: 512 MB

Given n integers, your task is to report for each integer the number of its divisors.

For example, if $x = 18$, the correct answer is 6 because its divisors are 1, 2, 3, 6, 9, 18.

Input

The first input line has an integer n : the number of integers.

After this, there are n lines, each containing an integer x .

Output

For each integer, print the number of its divisors.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq x \leq 10^6$

Example

Input	Output
3 16 17 18	5 2 6